

# FIELDS — Development Specification v1.2

**Working title:** Fields (placeholder) **Genre:** Cozy first-person mowing & haymaking sim  
**Engine:** Unity (URP) **Target length:** 2.5–3.5 hours to completion **Target price:** \$6.99–8.99 USD **Platform:** Windows (Steam), Steam Deck verified target **Document purpose:** Implementation spec + basis for fixed-price quotation

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## 0. HOW TO USE THIS DOCUMENT

This spec is written to be quoted in **two independently cancellable stages**:

- **Stage 1** — Full single-player game (milestones M1–M5)
- **Stage 2** — Co-operative multiplayer (2–4 players), added on top of Stage 1

Please quote each stage separately, and each milestone within Stage 1 separately. See Section 17 for the required quote format.

Section 16 (Out of Scope) is binding. Anything not listed in this document is not included.

**Section 8 (Game Feel) is the most important section in this document.** It is written as a set of testable requirements, not as flavour text. A build that implements every system in this spec but fails Section 8 is not an acceptable build.

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## 1. DESIGN PILLARS

1. **The tool is the game.** Everything lives or dies on how the cut feels — the sound, the sharp edge left behind, the rhythm. Content is secondary.
  2. **Visible progress.** The player must always be able to see how much is done and how much remains, without opening a menu.
  3. **Escalation you can feel.** Going from a hand sickle to a ride-on mower must feel like a transformation, not a stat increase.
  4. **Stylized, warm, low-spec.** Cartoon/stylized art, warm bright palette. Deliberately *not* a realistic simulator — this avoids competing with Lawn Mowing Simulator and avoids domain-expert criticism.
  5. **KISS.** No weather system, no crop growth simulation, no NPC schedules, no dialogue trees, no time-of-day progression. Every system in this document is intentionally minimal.
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## 2. CORE LOOP

CUT grass → hay accumulates → BALE it → TRANSPORT bale to stand → SELL → UPGRADE → next parcel

One loop iteration  $\approx$  90–180 seconds in early game,  $\approx$  45–60 seconds late game.

Cutting is the primary activity ( $\sim$ 65% of playtime), transport is the secondary rhythm-breaker ( $\sim$ 25%), menus and buying ( $\sim$ 10%).

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### 3. CAMERA & PLAYER REPRESENTATION

#### 3.1 View — CONFIRMED: FIRST-PERSON

The game is **first-person**. Only stylized hands/forearms and the active tool are rendered. There is no third-person camera in single-player.

#### 3.2 Player representation (single-player)

- **First-person camera.** Only the tool and a pair of stylized hands/forearms are rendered.
- **No full player body, no full character rig, no locomotion animation set.** This is a deliberate cost decision.
- Animation requirement is limited to: tool-specific hand+tool animations (idle, use, exhausted).

#### 3.3 Ride-on mower exception

When the player mounts the ride-on mower, the camera moves to a **seated first-person position** on the machine. Hands are visible on the steering wheel. No third-person orbit camera is required.

#### 3.4 Co-op exception

In co-op, *other* players must be visible as full characters. This requires:

- One simple stylized character model
- One shared rig
- A minimal animation set (idle, walk, run, use-tool, carry, push)
- Held-tool attachment sockets

This cost belongs to **Stage 2**, not Stage 1. Single-player must not require a full character.

#### 3.5 Controls

- WASD movement, mouse look
- LMB — use tool (hold to repeat)

- E — interact (mount mower, pick up bale, deposit at stand)
  - Q — drop carried bale
  - Tab — journal / progress
  - Scroll or 1-5 — tool select
  - Full gamepad support required (Steam Deck target)
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## 4. GRASS SYSTEM — CRITICAL PATH

This is the single highest technical risk in the project. Its acceptance criteria are binding at milestone M1 (Section 14.2).

### 4.1 Architecture

Grass state is **not** stored as objects. It is stored in two parallel representations:

#### A) Visual mask (GPU)

- One `RenderTarget` per field, format R8, 1024×1024 or 2048×2048 depending on field size
- White = uncut, black = cut
- Cutting writes a black circle/capsule into the mask
- Writes must use a min-blend or equivalent so already-cut areas cannot be "uncut"
- Ping-pong buffers or a `CommandBuffer` quad draw — **not** a self-referencing full-screen Blit

#### B) Logical grid (CPU)

- A `bool[ , ]` or `byte[ , ]` at much coarser resolution (~0.4 m per cell)
- Updated by the same cut operations
- This is the **source of truth** for: completion %, hay yield, save/load, and co-op sync
- The GPU mask is never read back to CPU

This separation is mandatory. Do not derive gameplay state from the render texture.

### 4.2 Rendering

- Grass is chunked (suggested 10×10 m chunks), each chunk one mesh with pre-generated blades
- Shader Graph reads the mask in the **vertex stage** (`Sample Texture 2D LOD`) and scales blade height by mask value
- **Cut grass must be flattened to stubble, not discarded.** The sharp visible boundary between cut and uncut is the core satisfaction of the game.
- Distance-based LOD; grass density falls off with distance

- `DrawMeshInstancedIndirect` only if profiling proves chunked meshes insufficient — do not start there

### 4.3 Cut operation API

```
CutArea(Vector3 worldPos, float radius)
CutCapsule(Vector3 from, Vector3 to, float radius)
```

Swing-based tools must sample along the blade arc across multiple frames and interpolate between frame positions, or fast swings will produce dotted/patchy cuts. Wheeled tools use `CutCapsule` from last frame position to current.

### 4.4 Performance targets (hard requirements)

| Metric           | Target                                                                        |
|------------------|-------------------------------------------------------------------------------|
| Frame rate       | 60 fps sustained on Intel Iris Xe / equivalent integrated GPU at 1080p Medium |
| Steam Deck       | 60 fps at 1280×800                                                            |
| Grass draw calls | < 150 in view                                                                 |
| Cut latency      | Visible change within 1 frame of input                                        |
| Memory           | < 2 GB                                                                        |

A build that does not hit the integrated-GPU target is not accepted. The target audience for this genre plays on laptops.

## 5. TOOLS

Five tools, acquired in fixed order. Each has **exactly three upgradeable stats**, each with **three purchasable levels** (base + 3 upgrades = 4 tiers per stat).

### 5.1 Stat design principle

Every tool uses the same three conceptual axes so the UI and the code are identical:

| Axis      | Manual tools              | Powered tools           |
|-----------|---------------------------|-------------------------|
| SPEED     | Swing speed               | Movement / cutting rate |
| ENDURANCE | Stamina before exhaustion | Fuel tank capacity      |

| POWER | Sharpness (passes to clear) | Cutting deck width |
|-------|-----------------------------|--------------------|
|-------|-----------------------------|--------------------|

5.2 Tool table

| # | Tool                                   | Unlock   | SPEED        | ENDURANCE     | POWER                                     |
|---|----------------------------------------|----------|--------------|---------------|-------------------------------------------|
| 1 | Hand Sickle (one-handed, short curved) | Start    | Swing speed  | Stamina pool  | Sharpness — passes needed to clear a cell |
| 2 | Long Scythe (two-handed)               | Purchase | Swing speed  | Stamina pool  | Cut arc width                             |
| 3 | String Trimmer (handheld)              | Purchase | Cutting rate | Fuel capacity | Head width                                |
| 4 | Push Mower                             | Purchase | Push speed   | Fuel capacity | Deck width                                |
| 5 | Ride-On Mower                          | Purchase | Drive speed  | Fuel capacity | Deck width                                |

5.3 Suggested values (tuning pass required)

| Tool        | Stat             | L0   | L1   | L2   | L3   |
|-------------|------------------|------|------|------|------|
| Sickle      | Swing time (s)   | 0.90 | 0.75 | 0.62 | 0.52 |
|             | Stamina (swings) | 20   | 28   | 38   | 50   |
|             | Passes to clear  | 3    | 2    | 2    | 1    |
| Long Scythe | Swing time (s)   | 1.10 | 0.92 | 0.78 | 0.66 |
|             | Stamina (swings) | 18   | 26   | 36   | 48   |
|             | Arc width (m)    | 1.4  | 1.7  | 2.0  | 2.4  |
| Trimmer     | Cut rate mult.   | 1.00 | 1.20 | 1.45 | 1.75 |
|             | Fuel (s of use)  | 90   | 130  | 180  | 240  |
|             | Head width (m)   | 0.5  | 0.6  | 0.7  | 0.85 |
| Push Mower  | Move speed (m/s) | 2.2  | 2.6  | 3.0  | 3.5  |
|             | Fuel (s of use)  | 150  | 210  | 280  | 360  |
|             | Deck width (m)   | 0.6  | 0.75 | 0.9  | 1.1  |

|         |                  |     |     |     |     |
|---------|------------------|-----|-----|-----|-----|
| Ride-On | Move speed (m/s) | 4.0 | 4.8 | 5.6 | 6.5 |
|         | Fuel (s of use)  | 240 | 330 | 440 | 570 |
|         | Deck width (m)   | 1.2 | 1.5 | 1.8 | 2.2 |

All values live in a ScriptableObject asset and must be editable without code changes.

### 5.4 Tool obsolescence — mandatory design rule

Later tools must **not** fully replace earlier ones, or the last hour of the game becomes monotonous. Because the map is a single meadow (Section 7.3), variety cannot come from changing biomes — it must come from **obstacle placement inside the meadow**.

- The ride-on mower cannot fit between trees, along fence lines, into corners, or onto steep banks
- The push mower cannot reach tight edges around posts, rocks, troughs and walls
- The trimmer and scythes handle all edge work and all awkward pockets

**Mandatory obstacle budget per parcel** (parcels 2-4), supplied by client as environment art:

| Obstacle                     | Purpose                                | Min. count |
|------------------------------|----------------------------------------|------------|
| Scattered trees / stumps     | Force tool switching around trunks     | 6          |
| Fence line segments          | Long edges only the trimmer can finish | 40 m       |
| Rocks / boulders             | Obstacle strikes, edge work            | 8          |
| Posts, troughs, gates, wells | Tight pockets                          | 5          |
| Ditch or bank (8-12°)        | Ride-on cannot climb, adds slope       | 1          |

**Target: at least 12% of every parcel's cuttable area must be unreachable by the ride-on mower.** This is a level design requirement and a testable acceptance criterion, not a suggestion.

### 5.5 Stamina and fuel

- **Manual tools:** stamina bar drains per swing, regenerates when not swinging. At zero, swing speed drops ~60% until partially recovered. It never hard-blocks input.
- **Powered tools:** fuel drains while running. Refuel free at the shed/stand — no fuel purchasing, no fuel economy. Fuel exists only to pace the session and make the ENDURANCE upgrade meaningful.

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## 6. HAY & BALING

### 6.1 Hay accumulation — "the field remembers"

Hay is **not** an abstract meter. It is a physical object that stays exactly where the player cut it. This is the system that turns cutting into visible labour and creates the game's second-pass rhythm.

#### 6.1.1 Accumulation rule

- Every logical grid cell (Section 4.1B) that transitions from uncut to cut yields **1 hay unit**
- Hay units accumulate into the nearest **collection cell** — a coarse grid over the map, suggested 6×6 m
- When a collection cell reaches **60 hay units**, a **Hay Pile** object spawns at the weighted centroid of the cells that contributed to it, and the collection cell resets to zero
- Leftover units below the threshold persist and carry over — nothing is ever lost
- A partially filled collection cell shows a loose-hay decal that grows through three stages, so the player can see a pile is about to form

**Result:** cutting a parcel leaves piles distributed along the exact path the player took. The field becomes a map of the player's own work, and two players cutting the same parcel produce visibly different layouts.

#### 6.1.2 Pile behaviour

- Piles are static props with a collider; three size variants for visual variety
- Piles are **carried**, reusing the same carry code and the same carry-capacity stat as square bales (Section 6.2)
- Piles never decay, never move, and are affected by nothing
- Piles must read clearly against cut stubble from across a parcel

#### 6.1.3 Conversion at the baler

| Product     | Cost        | Notes                          |
|-------------|-------------|--------------------------------|
| Square bale | 1 Hay Pile  | Default, available from start  |
| Round bale  | 3 Hay Piles | Requires purchased round baler |

Depositing a pile is instant; the baler then runs its compression cycle (Section 6.2) and ejects the finished bale.

### 6.1.4 Why this design

- The volume of work becomes **visible in the world** instead of being a HUD number
- It creates a natural rhythm break — cut a section, then do a gathering run — without adding a new tool or a new verb
- It gives carry capacity a second, earlier purpose (piles, not just bales), so that upgrade pays off from minute five
- It explains the round bale cleanly: three piles, one object, one trip
- It requires no new systems; piles reuse the bale carry code entirely

### 6.1.5 Parcel completion

A parcel counts as complete only when **grass is 100% cut AND zero Hay Piles remain in it**. Leaving hay behind is the game's only form of unfinished business, and it is self-evident on sight — no checklist required.

## 6.2 Baler

One baler machine (static), placed at the farmstead beside the stand. There is only one in the game.

### Square bale — default, available from start

| Stat              | Description                                  | L0    | L1    | L2    | L3    |
|-------------------|----------------------------------------------|-------|-------|-------|-------|
| Compression speed | Time to form one bale (s)                    | 6.0   | 4.5   | 3.4   | 2.5   |
| Carry capacity    | <b>Piles or square bales</b> carried at once | 1     | 2     | 3     | 4     |
| Density           | Hay per bale → value per bale                | 1.00× | 1.15× | 1.30× | 1.50× |

### Round bale — must be purchased

| Stat              | Description                          | L0    | L1    | L2    | L3    |
|-------------------|--------------------------------------|-------|-------|-------|-------|
| Compression speed | Time to form one bale (s)            | 9.0   | 7.0   | 5.5   | 4.0   |
| Roll control      | Push speed / steering responsiveness | 1.00× | 1.20× | 1.40× | 1.65× |
| Density           | Hay per bale → value per bale        | 2.50× | 2.85× | 3.25× | 3.75× |

Round bales hold three piles each, are slower to make, and can only be moved by rolling.

**Balancing intent:** at base density a round bale is worth slightly less than three square bales. The round baler is a *convenience* purchase — one object, one trip, no carry speed penalty — that becomes strictly better only after its Density upgrades. This is deliberate:



the player should feel they made the round baler good, rather than being handed it.

6.3 Transport — the second core mechanic

**Square bales:** picked up and carried in first person. Carrying reduces movement speed ~25%. Carrying 2+ reduces it further and partially obstructs the lower screen. Bales are placed with E, dropped with Q.

**Round bales:** pushed/rolled with physics, for the entire game. **CONFIRMED: there is no trailer, hitch or vehicle transport for round bales at any point.** Rolling by hand is the intended experience from first purchase to credits. This is the game's signature mechanic and must feel good:

- Rigidbody with tuned mass, drag and angular drag
- Rolls downhill on its own; the player must manage that, not fight it
- Collides with fences, trees, rocks, and other bales
- Can knock over light props (buckets, signs, small crates) — **this is intentional and must not be removed**
- Must not be able to escape the field bounds or clip through terrain

Physics tuning for the round bale is a named deliverable, not an afterthought.

6.4 Selling

The **Stand** is a fixed structure at each field. Depositing a bale sells it instantly and pays out. No inventory, no market prices, no contracts, no haggling.

Hay Value — 3 upgrade levels

| Level | Price multiplier | Cost   |
|-------|------------------|--------|
| Base  | 1.00×            | —      |
| I     | 1.25×            | tuning |
| II    | 1.55×            | tuning |
| III   | 1.90×            | tuning |

This is a single global multiplier applied to all bale sales.

7. ECONOMY & PROGRESSION

7.1 Currency

One currency. Earned only from selling bales. Spent only in the shop.

## 7.2 Shop

Accessible at the stand. Flat list, three tabs:

- **Tools** — purchase the 5 tools in order
- **Upgrades** — 3 stats × 3 levels for each owned tool, plus baler stats
- **Unlocks** — round baler, next parcel

No shopkeeper NPC, no dialogue, no shop building interior. A UI panel at the stand.

## 7.3 Map structure — CONFIRMED

**One continuous map. One biome: meadow. Four parcels of increasing size, unlocked by purchase.**

There are no separate levels, no loading between parcels, and no environment theme changes. The map is a single meadow farmstead. Parcels are divided by fence lines and gates; buying a parcel opens its gate.

| Parcel            | Relative size | Cuttable area         | Introduces                             | Target time |
|-------------------|---------------|-----------------------|----------------------------------------|-------------|
| 1 — Home Paddock  | 1×            | ~1,200 m <sup>2</sup> | Sickle, square baler, stand, hay piles | 25-35 min   |
| 2 — Middle Meadow | 2×            | ~2,400 m <sup>2</sup> | Long scythe, trimmer                   | 35-45 min   |
| 3 — Far Meadow    | 3.5×          | ~4,200 m <sup>2</sup> | Push mower, round baler                | 45-60 min   |
| 4 — Top Meadow    | 5×            | ~6,000 m <sup>2</sup> | Ride-on mower                          | 50-70 min   |

Total ≈ 2.5-3.5 hours.

### 7.3.1 Single farmstead — a deliberate pacing device

There is exactly **one baler and one stand**, both at the farmstead adjacent to Parcel 1. All hay and all bales must come back to that single point.

Haul distance therefore grows with every parcel unlocked, which does three things for free:

- Makes the carry-capacity upgrade progressively more valuable
- Makes the round bale progressively more attractive — roll it, no carry speed penalty, over the longest distances
- Gives the map a centre of gravity and a sense of home base without needing a house interior or any additional art

### 7.3.2 Terrain

- Parcels 1-2: flat, gentle undulation only
- Parcel 3: one bank or ditch at 8-12°
- Parcel 4: rolling terrain with sustained slopes, maximum gradient **12°**

**12° is the confirmed maximum gradient.** The ride-on mower must handle it without feeling unstable, and round bales must roll away convincingly on it while never gaining enough speed to escape map bounds. Parcel 4's slopes exist specifically to make round-bale rolling dramatic on the longest hauls in the game.

### 7.4 Completion

A parcel is complete when grass is 100% cut and no Hay Piles remain in it (Section 6.1.5). Completing all four parcels triggers a short end screen with total stats — bales made, hay sold, time taken, completion per parcel. No cutscene, no narrative.

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## 8. GAME FEEL — THE MOST IMPORTANT SECTION

This game has no story, no combat, no difficulty curve and no social systems. Everything the player buys the game for happens in the 200 milliseconds around a single swing, and in the physics of a bale going down a hill. If those are excellent, a thin game succeeds. If they are merely functional, no amount of content saves it.

This section is written as **testable requirements**. Every numbered item is checkable in a build.

### 8.0 The governing rule

Every player action must produce a change in the world within one frame, and that change must be audible, visible, and felt in the camera.

No action in this game may resolve silently. No action may resolve only visually. No action may be delayed by an animation the player did not ask for.

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### 8.1 Manual scything — the core verb

The sickle and long scythe carry the first 40 minutes of the game. They must be the most polished thing in the build.

#### 8.1.1 Swing cycle structure

A swing is three phases with fixed proportions, scaled by the tool's SPEED stat:

| Phase | Proportion | Behaviour |
|-------|------------|-----------|
|-------|------------|-----------|

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|          |     |                                                                                         |
|----------|-----|-----------------------------------------------------------------------------------------|
| Wind-up  | 25% | Tool pulls back. Cut is NOT active. Camera drifts ~1° opposite the swing direction.     |
| Sweep    | 30% | Cut IS active across the whole phase. This is where the arc is sampled.                 |
| Recovery | 45% | Tool returns. Cut is NOT active. Next queued swing may begin at 70% through this phase. |

At base sickle speed (0.90 s cycle): wind-up 0.22 s, sweep 0.27 s, recovery 0.41 s.

8.1.2 Arc sampling (mandatory)

The cut must be sampled from the blade tip transform on **every frame of the sweep phase**, and consecutive frame positions must be interpolated with `CutCapsule` so no gaps appear.

- Interpolation step: never larger than 50% of the cut radius
- Test: at maximum SPEED upgrade, with mouse look moving at maximum sensitivity, the cut boundary must contain zero gaps
- The arc must follow the actual blade mesh, not a fixed offset from the camera. Looking down cuts closer; looking level cuts further.

8.1.3 Hit vs. whiff — mandatory distinction

The game must know, per swing, how much uncut grass the arc actually intersected, and respond differently:

| Result   | Condition                               | Audio                          | Camera                               | Particles            |
|----------|-----------------------------------------|--------------------------------|--------------------------------------|----------------------|
| Full hit | >60% of arc cells were uncut            | Full crunch layer, loudest     | Kick 2.0°                            | Full burst           |
| Partial  | 15-60%                                  | Mid layer, -3 dB               | Kick 1.2°                            | Reduced burst        |
| Whiff    | <15%                                    | Air-swish only, no crunch      | Kick 0.6°                            | None                 |
| Obstacle | Arc intersects rock/fence/post collider | Sharp metallic clang, distinct | Kick 3.0°, <b>reversed direction</b> | Sparks + tool bounce |

The whiff sound is what makes the hit sound feel good. Do not skip it.

The obstacle hit must physically interrupt the swing animation — the tool bounces back. This single detail does more for perceived weight than any other item in this

section.

#### 8.1.4 Rhythm and input

- Holding LMB repeats swings indefinitely at a **metronomic, unchanging interval**. No randomization of timing.
- Swing direction alternates left→right→left automatically. The player never chooses direction.
- A new swing input during recovery is **queued**, never dropped. Input must never be swallowed by an animation.
- The interval is the game's heartbeat. It must be stable enough that a player can fall into it without thinking. This is the meditative quality the genre sells.

#### 8.1.5 Stamina feel

- Stamina drains per swing, not per second
- Regeneration begins 1.2 s after the last swing
- At zero stamina: swing cycle time increases 60%, audio gains a breath layer, camera bob increases slightly, and the tool visibly droops in the idle pose
- **Stamina never blocks input**. Exhaustion slows the player down; it never stops them. A hard block would break the meditative loop and feel punitive.
- Recovery to usable (30%) must take no longer than 4 seconds

#### 8.1.6 The first-cut moment

The very first swing of a new game must be the strongest single sensory moment in the first ten minutes. Full-hit audio, generous particles, and a slightly exaggerated camera kick (3°) on the first three swings only, then settling to normal values. The player must understand within one second that this game is about the feeling of cutting.

#### 8.1.7 The sickle → long scythe transition

When the player buys the long scythe, the difference must be immediately legible without reading stats:

- Wider arc (visible as a much broader cut swath)
- Slower, heavier cycle
- Lower-pitched, longer audio with more low-end
- Larger camera movement
- The player should feel they are now using a real tool, not a hand implement

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### 8.2 Powered hand tools

#### 8.2.1 String trimmer

- Continuous cut while held, not discrete swings
- Engine audio has three layers: idle loop, revving transition, load loop
- **Bogging:** when the head is inside dense uncut grass, RPM audio drops in pitch by ~15% and cut rate reduces ~20%. When it exits into cut ground, RPM rises again. This one behaviour is what makes a trimmer feel like a trimmer.
- Continuous small camera vibration (0.3° amplitude, ~18 Hz) while running
- Controller: light continuous rumble while running, stronger during bogging

### 8.2.2 Push mower

- Cut is continuous from a `CutCapsule` swept between last-frame and current-frame deck positions
  - **Movement resistance:** pushing into uncut grass reduces movement speed by 20%; pushing over already-cut ground is full speed. The player physically feels the work.
  - Engine bogging as above, tied to the volume of grass being cut per second
  - Wheels cast dust puffs on dry ground and grass flecks on cut ground
  - Turning has a visible pivot around the rear wheels, not instant rotation
- 

### 8.3 Ride-on mower — the payoff vehicle

This is the reward for two hours of manual labour. It must feel like a reward.

- **Arcade handling only.** No WheelColliders, no realistic suspension, no simulation of weight transfer. A kinematic or simple rigidbody controller with tuned acceleration curves.
  - Acceleration: 0 → full speed in ~1.5 s, with audible engine ramp
  - Braking: firm, ~0.8 s to stop
  - Steering: speed-sensitive — tighter turning at low speed, wider at high speed
  - Visible body roll on turns (max 4°) and pitch on acceleration/braking (max 3°)
  - Seated camera has gentle suspension bounce over uneven terrain
  - **Deck engage/disengage is a separate input.** Engaging the deck produces a satisfying mechanical clunk and a drop in engine pitch as the blades load. Driving with the deck off is faster and quieter.
  - A continuous, wide, uninterrupted cut swath behind the machine — the visual signature of the late game
  - Grass discharge particles from the side of the deck
  - The first 30 seconds of driving this machine should make the player laugh at how fast the field disappears
-

## 8.4 Bale formation

- Depositing hay into the baler starts a **visible mechanical process**, not an instant spawn
  - The baler shakes, the audio builds, a progress indication is diegetic (a moving arm, a filling chamber — client's environment art)
  - Compression completes with a hard mechanical **thunk** and a slight camera shake if the player is nearby
  - The bale is **ejected with physics**, not placed. It drops, bounces once, settles.
  - A puff of dust and loose hay particles on ejection
  - Round bale ejection is slower, heavier, lower-pitched, and shakes the ground more
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## 8.5 Square bale carrying

- Pickup has a short animation and a grunt-adjacent audio cue (non-verbal, stylized)
  - Carrying one bale: movement speed –25%, subtle vertical bob synced to footsteps, bale visible in the lower portion of the screen
  - Carrying two: speed –40%, larger bob, bale partially obstructs the lower screen, footstep audio becomes heavier
  - Carrying three: speed –50%, view obstruction significant enough that the player must plan routes
  - **The obstruction is the design.** Carrying more is faster in total but harder to navigate. This is the tension that makes the capacity upgrade a real decision.
  - Dropping a bale is instant, physical, and produces a dull thud with a small dust puff
  - Bales stack on each other and can be knocked over
- 

## 8.6 Round bale rolling — the signature mechanic

**This is the mechanic people will clip and share. It gets its own tuning pass and its own acceptance test.**

### 8.6.1 Physical behaviour

- Rigidbody, mass tuned so it feels heavy but never immovable
- **Inertia is the point.** It takes ~1 second of pushing to get moving and ~1.5 seconds to stop. The player manages momentum rather than steering precisely.
- Rolls freely downhill under gravity. On slopes above ~8°, it will get away from the player. **This is intentional and must not be "fixed."**
- Angular drag tuned so it rolls convincingly on grass and faster on cut stubble
- Pushing from an off-centre position causes it to curve — steering is done by

choosing where you push

### 8.6.2 Interaction with the world

- Collides with and knocks over: buckets, signs, small crates, fence posts, tool racks, other bales
- Fences and walls stop it with a heavy impact sound and visible deformation of nothing (no destruction system) — the impact is sold entirely by audio and camera shake
- Can knock the player over? **No** — see anti-patterns below. It pushes the player aside.
- Cannot leave field bounds or clip through terrain under any circumstance. This is a hard bug bar.

### 8.6.3 Audio

- Continuous rolling loop, pitch and volume scaled to angular velocity
- Surface-dependent: uncut grass (muffled, soft), cut stubble (rougher, crisper), dirt (dull), gravel (harsh)
- Impact sounds scaled to collision force, with a distinct heavy variant above a threshold
- A low-frequency rumble layer when the bale is very close to the camera

### 8.6.4 Camera

- Slight camera shake proportional to rolling speed when within 3 m
- No camera lock-on, no automatic framing. The player looks where they want.

### 8.6.5 Acceptance test

Place a round bale at the top of the steepest slope in Field 4. Release it. If watching it go is not entertaining on its own, without any goal attached, the tuning is not finished.

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## 8.7 Selling

- Delivering a bale to the stand triggers an immediate, generous payout sound — layered coin/register audio, bright and short
  - Money counter animates upward over ~0.4 s rather than snapping
  - A brief, small particle flourish at the stand
  - Multi-bale deliveries chain the sound with rising pitch per bale (max 5 steps) — this makes hauling three bales at once feel earned
-



## 8.8 Audio specification

### 8.8.1 Variation requirements

| Sound                   | Min. variations | Pitch jitter            |
|-------------------------|-----------------|-------------------------|
| Sickle full hit         | 5               | ±5%                     |
| Sickle whiff            | 3               | ±7%                     |
| Long scythe full hit    | 5               | ±5%                     |
| Obstacle strike         | 3               | ±4%                     |
| Footstep (per surface)  | 6               | ±8%                     |
| Bale drop               | 4               | ±6%                     |
| Bale impact             | 5               | scaled by force         |
| Bale formation complete | 2               | ±3%                     |
| Sale payout             | 3               | pitch-stepped on chains |

Repetition is the single most common failure mode in this genre. A player will hear the cut sound several thousand times per playthrough.

### 8.8.2 Mixing

- Audio mixer with separate groups: Master, Tools, World, Ambience, UI, Music
- Tool sounds duck ambience by ~3 dB during activity
- Engine loops must crossfade between idle/rev/load without audible seams
- No sound may clip at maximum simultaneous activity (ride-on running + bale rolling + ambience)
- Distance attenuation on all world sounds; the baler must be audible across a field but not dominant

### 8.8.3 Music — CONFIRMED: ambient only

There is **no composed musical score**. The game ships with ambient soundscapes only.

- Layered ambient beds: wind through grass, birds, insects, distant livestock, occasional distant bell or engine
- Layers must be randomized so no loop point is audible across a 60-minute session
- Ambience must thin out to near-silence for long stretches — silence plus wind is a valid and desirable state in this genre
- A single soft ambient tone may fade in on parcel completion as a reward cue: this is

A single soft ambient tone may fade in on parcel completion as a reward cue, this is the only non-diegetic audio in the game

- Ambience always sits below tool audio in the mix

This removes the composer line item entirely and is a deliberate cost and tone decision.

---

## 8.9 Camera specification

| Parameter           | Value                                       |
|---------------------|---------------------------------------------|
| Default FOV         | 70° (slider 60–100)                         |
| Head bob amplitude  | 0.03 m vertical, 0.015 m lateral            |
| Head bob frequency  | Synced to footstep rate                     |
| Head bob toggle     | Required in settings (accessibility)        |
| Swing camera kick   | 0.6–3.0° per Section 8.1.3                  |
| Kick recovery       | Spring-damped, ~0.25 s to rest              |
| Landing impact      | 2° dip, 0.2 s recovery                      |
| Camera shake toggle | Required in settings, with intensity slider |

All camera movement must be disableable. Some of this audience is motion-sensitive, and a "makes me nauseous" review in a cozy game is disproportionately damaging.

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## 8.10 Haptics

Gamepad rumble is required and must be subtle:

- Swing full hit: short sharp pulse, 0.08 s
  - Whiff: no rumble
  - Obstacle strike: sharp double pulse
  - Trimmer/mower running: continuous low-amplitude
  - Bogging: amplitude increase
  - Round bale rolling nearby: low continuous, scaled by proximity and speed
  - Bale impact: single strong pulse
- 

## 8.11 Anti-patterns — explicitly forbidden

These are the specific ways this genre is usually ruined:

1. **Cut animations that block input.** Never. Queue instead.
  2. **Randomized swing timing.** Destroys the meditative rhythm.
  3. **Hard stamina lockout.** Slow the player, never stop them.
  4. **A single cut sound with no variation.** The fastest route to a mediocre review.
  5. **Discarded grass instead of stubble.** The sharp cut boundary is the product.
  6. **Realistic vehicle physics.** Frustrating, expensive, and wrong for the tone.
  7. **The player being knocked down or damaged by bales.** There is no fail state in this game. Bales push the player aside; they never punish.
  8. **Fixing the round bale rolling away downhill.** That is the game.
  9. **Snapping numbers in the UI.** Animate money and progress values.
  10. **Tutorial popups.** Maximum 5 one-line contextual hints in the entire game.
- 

## 8.12 Acceptance tests for Section 8

The following must all pass before Stage 1 is considered complete:

1. **The 60-second test.** A new player is given the sickle and no instructions. After 60 seconds of cutting, they are still cutting without being asked to.
  2. **The silent test.** With all audio muted, cutting still reads clearly through camera and particles alone. With the screen off, cutting still reads clearly through audio alone. Both channels must independently carry the action.
  3. **The gap test.** Maximum-speed swinging with maximum mouse movement leaves zero gaps in the cut boundary.
  4. **The hill test.** Section 8.6.5.
  5. **The repetition test.** Ten minutes of continuous cutting without the audio becoming annoying.
  6. **The transition test.** Buying each new tool produces an audible and visible "this is better" moment within the first three seconds of use.
- 

## 9. UI

Minimal, diegetic where possible.

HUD:

- Stamina/fuel bar (only visible for the active tool, fades out when full)
- Carried hay meter
- Carried bale indicator

- Money counter (fades in on change)
- Field completion % (top corner, small)

## Screens:

- Shop panel
- Journal/progress (Tab): per-field completion, bales made, money earned, tool levels
- Settings: resolution, quality preset, volume sliders, sensitivity, invert Y, key rebinding, language
- Main menu, pause menu

No minimap. No quest log. No tutorial popups beyond 4-5 one-line contextual hints.

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## 10. SAVE SYSTEM

- Single save slot, autosave on: field completion, purchase, bale sold, and every 60 s
  - Must persist: money, owned tools, all upgrade levels, unlocked fields, per-field cut grid, bales currently in the world (position + type)
  - The cut grid is saved from the **CPU logical grid**, run-length encoded or bit-packed — not as an image
  - Save file must survive version updates gracefully (versioned schema)
- 

## 11. STEAM INTEGRATION

- Steamworks SDK integration
  - **25 achievements** (list to be supplied by client)
  - Steam Cloud save
  - Steam Deck input support and verification checklist
  - Rich Presence: current field name
- 

## 12. LOCALIZATION

- All strings externalized from day one (no hardcoded UI text)
- Ship languages: EN, HU, DE, RU, ZH-Hans, PL, ES, PT-BR, JP
- Translation is supplied by client; vendor implements the system and integrates the files
- UI must handle 2× string length expansion without breaking layout

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### 13. ART RESPONSIBILITY SPLIT

| Asset                                                                  | Provider                                                 |
|------------------------------------------------------------------------|----------------------------------------------------------|
| All environment art (terrain, buildings, fences, trees, props, skybox) | Client                                                   |
| Grass mesh + material                                                  | Vendor (technical)                                       |
| 5 tool models                                                          | Client                                                   |
| First-person hands model + tool animations                             | To be confirmed — please quote separately                |
| Bale models (square, round)                                            | Client                                                   |
| VFX (particles, dust, trails)                                          | Vendor                                                   |
| UI art and icons                                                       | To be confirmed — please quote separately                |
| Audio (SFX + ambience + music)                                         | To be confirmed — please quote separately                |
| Co-op character model + animations                                     | To be confirmed — please quote separately (Stage 2 only) |

Please price the "to be confirmed" rows as optional line items so they can be included or excluded.

---

### 14. STAGE 1 — SINGLE PLAYER (quote separately)

Everything in Sections 3-13.

#### 14.1 Milestones

| #  | Milestone      | Contents                                                                                                                                                    |
|----|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1 | Vertical slice | Field 1 complete: grass system, sickle, hay, square baler, carry, stand, shop, save. Section 8 fully applied to the sickle. Playable and shippable in feel. |
| M2 | Tools complete | All 5 tools with all upgrade stats, stamina/fuel, tool switching, Section 8.2 and 8.3 applied                                                               |

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|    |                  |                                                                                                                         |
|----|------------------|-------------------------------------------------------------------------------------------------------------------------|
| M3 | Baling complete  | Round baler, roll physics tuned per Section 8.6, transport, all economy upgrades                                        |
| M4 | Content complete | All 4 fields playable, full progression curve tuned, all UI                                                             |
| M5 | Ship candidate   | Steam integration, achievements, localization system, settings, Deck support, full audio pass, optimization, bug fixing |

### 14.2 M1 acceptance criteria — binding

M1 is the project's decision gate. It carries the grass system's technical risk and the game's entire feel proposition. All of the following must pass before M1 is signed off and M2 begins:

#### Technical

1. Sustained 60 fps at 1080p on an integrated GPU (Intel Iris Xe or equivalent)
2. Sustained 60 fps at 1280×800 on Steam Deck
3. No frame hitching during cut operations
4. Cut boundary sharp and gap-free at maximum swing speed with maximum mouse movement
5. CPU completion percentage matches visual state within 2%
6. Cut state, money, and world bales survive save/load correctly
7. Memory under 2 GB

**Feel** 8. Acceptance tests 1, 2, 3 and 5 from Section 8.12 all pass 9. Hit / partial / whiff / obstacle distinction implemented per Section 8.1.3 10. Minimum audio variation counts from Section 8.8.1 met for all sickle sounds 11. Camera and haptic response implemented per Sections 8.9 and 8.10

Deliverable: Windows build, Steam Deck build, and full Unity project source.

If M1 fails these criteria, the project stops. Vendors should price M1 with that in mind — it is the milestone that must be right.

## 15. STAGE 2 — CO-OP (quote separately)

Do not build this into Stage 1. Do build Stage 1 in a co-op-aware way.

### 15.1 Architecture requirement for Stage 1

Even in the single-player-only build, all mutable game state must be routed through a single authority object rather than scattered across MonoBehaviours reading each other directly. Single-player is treated as a session with one participant. This means very little

directly. Single-player is treated as a session with one participant. This costs very little now and prevents a rewrite later.

## 15.2 Co-op scope

- **2-4 players**, host-authoritative
- Steam lobbies, invite via Steam overlay, drop-in/drop-out
- Suggested stack: Mirror or FishNet + Steam transport (vendor may propose alternative)

## 15.3 Systems requiring network work

| System            | Approach                                                                                                                                |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Grass state       | Sync <b>cut events</b> (position, radius, tool), not the mask. All clients paint locally. Late joiners receive the compressed CPU grid. |
| Bale physics      | Server-authoritative. No client-side prediction — bales are slow, latency is acceptable. Ownership transfers on grab.                   |
| Economy           | Shared wallet. Purchases resolved by host. Handle simultaneous-purchase race conditions.                                                |
| Tools             | Each player has independent tool ownership and upgrade state, or shared — <b>client decision required, see Section 18</b>               |
| Save              | Host owns the save. Guest progress is not persisted.                                                                                    |
| Player characters | Requires the character model, rig, and animation set described in 3.4                                                                   |

## 15.4 QA burden

Co-op roughly doubles test effort and requires multi-machine testing. Please include this in the Stage 2 quote explicitly rather than folding it into development hours.

## 15.5 Cheaper alternative to price alongside

**Steam Remote Play Together** support (no netcode; Steam streams input from a second player). Please quote this as a separate low-cost line item. It delivers a meaningful share of the co-op value for a fraction of the cost and can ship at launch.

## 16. OUT OF SCOPE

Explicitly excluded. Any of these entering the project is a change request with separate pricing.

- Weather system, day/night cycle, seasons
  - Grass regrowth over time
  - Crop growth simulation, planting, fertilizing
  - Animals or livestock
  - NPCs, dialogue, quests, story
  - Vehicle damage, maintenance, repair
  - Realistic vehicle physics (arcade handling only)
  - Fuel purchasing or fuel economy management
  - Dynamic market pricing, contracts, reputation
  - Character customization or cosmetics
  - Terrain deformation
  - Photo mode
  - Mod support
  - Console ports, macOS, Linux native builds
  - Procedural field generation
  - Leaderboards
- 

## 17. QUOTE REQUEST FORMAT

Please respond with:

**A. Stage 1 — Full single-player** Fixed price per milestone M1-M5, total timeline, team composition and role hours. M1 must be priced and contractable on its own.

**B. Stage 2 — Co-op (2-4 players)** Fixed price, timeline, netcode stack proposed, QA hours listed separately.

### C. Optional line items

- First-person hands + tool animations
- UI art and icons
- Audio package
- Co-op character model + animations
- Steam Remote Play Together support

**D. Team** Please state specifically who will implement the grass system and what comparable work they have shipped. This is the role where experience matters most; a generalist gameplay developer is not sufficient.

**E. Assumptions and risks** Anything in this document you consider underspecified or risky.



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## 18. DECISIONS — RESOLVED

All previously open design decisions are closed. Recorded here so that vendors quote against fixed requirements.

| # | Decision             | Resolution                                                                                                             |
|---|----------------------|------------------------------------------------------------------------------------------------------------------------|
| 1 | Camera               | <b>First-person.</b> Hands and tools only. No full player body in single-player.                                       |
| 2 | Co-op tool ownership | <b>Individual per player.</b> Each player owns and upgrades their own tools. Shared wallet, separate tool progression. |
| 3 | Round bale transport | <b>Rolling only, for the entire game.</b> No trailer, no hitch, no vehicle transport.                                  |
| 4 | Achievements         | <b>25.</b> List supplied by client.                                                                                    |
| 5 | Music                | <b>Ambient only.</b> No composed score.                                                                                |
| 6 | Maximum gradient     | <b>12°</b> , Parcel 4 only.                                                                                            |
| 7 | Map structure        | <b>One map, one biome (meadow), four parcels</b> of increasing size. Single shared baler and stand at the farmstead.   |
| 8 | Hay accumulation     | <b>Physical Hay Piles</b> spawned in the field where cut. See Section 6.1.                                             |

### 18.1 Remaining client inputs (not decisions)

- 25-entry achievement list
- Environment art meeting the obstacle budget in Section 5.4
- Localization string files
- Tool models, hands model, bale and hay pile models

### 18.2 Note on individual co-op tool ownership

Individual ownership is the more satisfying choice and is confirmed, but vendors should be aware it carries real cost in Stage 2:

- Per-player inventory, upgrade state, stamina and fuel must all replicate
- The shop UI must resolve who is buying, from a shared wallet
- Simultaneous-purchase race conditions must be handled
- Late joiners start with base tools and cannot be brought to parity, so the host's session must remain playable for a fresh joiner

Please price this explicitly rather than folding it into general netcode hours.

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## 19. VERSION HISTORY

| Version | Date       | Notes                                                                                                                                                                                                                                                          |
|---------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 2026-07-21 | Initial specification                                                                                                                                                                                                                                          |
| 1.1     | 2026-07-21 | Stage 0 removed; its acceptance criteria folded into M1. Section 8 (Game Feel) expanded into a full testable specification.                                                                                                                                    |
| 1.2     | 2026-07-21 | All open decisions resolved. First-person confirmed. Map restructured to a single meadow with four parcels. Hay Pile accumulation system designed (6.1). Round-bale rolling-only confirmed. 25 achievements, ambient-only audio. Obstacle budget added to 5.4. |